

5 Performance Indicators

Performance indicators are used to judge the quality of a control system.

5.1 1st-order System Generic Performance Indicators

Generic performance indicators are:

- steady-state value $x_{ss} = \lim_{t \rightarrow \infty} x(t)$
- steady-state error $e_{ss} = \lim_{t \rightarrow \infty} e(t)$ where $e(t) = f(t) - x(t)$

To expand on the error definition, $e(t) = f(t) - x(t)$, consider a step response where $x(\infty) = 1$ and $f(\infty) = 1$, therefore $e(\infty) = 0$.

For a 1st-order system, x_{ss} and e_{ss} can be found using the final value theorem and exist in both the time domain and the s-domain. Recall the final value theorem for x_{ss} and e_{ss} leads to

$$x_{ss} = \lim_{s \rightarrow 0} sX(s) \quad (1)$$

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) \quad (2)$$

Therefore, starting at the transfer function of a 1st-order system

$$G(s) = \frac{1}{Ts + 1} \quad (3)$$

we can expand on this to show

$$\begin{aligned} X(s) &= G(s)F(s) \\ &= \frac{1}{Ts + 1} F(s) \end{aligned} \quad (4)$$

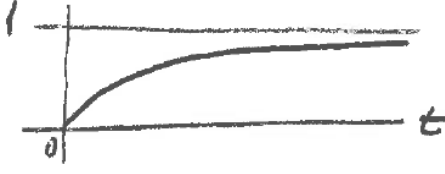
next, the error in s-domain is shown to be

$$E(s) = F(s) - X(s) \quad (5)$$

$$\begin{aligned} &= F(s) - G(s)F(s) \\ &= (1 - G(s))F(s) \\ &= \frac{Ts}{Ts + 1} F(s) \end{aligned} \quad (6)$$

Again, these are general terms for a 1st-order system.

5.1.1 Step response 1st-order system performance indicators



For a 1st-order system subjected to a step response, we want to find x_{ss} and e_{ss} and we know the transfer function is defined as

$$G(s) = \frac{1}{Ts + 1} \quad (7)$$

while the equation of motion is

$$T\dot{x} + x = F(s) \quad (8)$$

Where the step function is defined as

$$f(t) = 1, t > 0 \quad (9)$$

therefore, the system response is

$$x(t) = 1 - e^{-t/T} \quad (10)$$

while the steady-state response is

$$\begin{aligned} x_{ss} &= \lim_{t \rightarrow \infty} \\ &= 1 \end{aligned} \quad (11)$$

Next, the error as a function of time is

$$\begin{aligned} e(t) &= 1 - (1 - e^{-t/T}) \\ &= e^{-t/T} \end{aligned} \quad (12)$$

while the steady-state error is

$$\begin{aligned} e_{ss} &= \lim_{t \rightarrow \infty} e(t) \\ &= \lim_{t \rightarrow \infty} e^{-t/T} \\ &= 0 \end{aligned} \quad (13)$$

These same solutions can be found in the s-domain where the step function is $F(s) = \frac{1}{s}$. Consider the s-domain expression solved for $X(s)$,

$$X(s) = \frac{1}{Ts + 1} \cdot \frac{1}{s} \quad (14)$$

the steady-state error is shown to be

$$\begin{aligned} x_{ss} &= \lim_{s \rightarrow 0} s \frac{1}{Ts + 1} \cdot \frac{1}{s} \\ &= \lim_{s \rightarrow 0} \frac{1}{Ts + 1} \\ &= 1 \end{aligned} \quad (15)$$

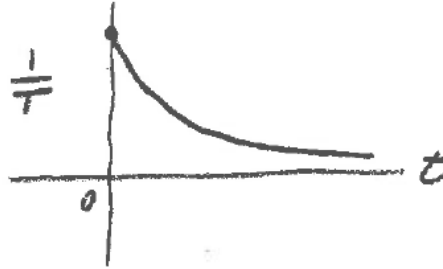
Next, we can build the s-domain representation of the error as

$$\begin{aligned} E(s) &= \frac{Ts}{Ts + 1} \cdot \frac{1}{s} \\ &= \frac{T}{Ts + 1} \end{aligned} \quad (16)$$

solving for the steady-state error results in

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s \frac{T}{Ts + 1} \\ &= 0 \end{aligned} \quad (17)$$

5.1.2 Impulse response 1st-order system performance indicators



For a 1st-order system subjected to a impulse response, we want to find x_{ss} and e_{ss} . The impulse function is defined as

$$f(t) = \delta(t) \quad (18)$$

therefore, the response is

$$x(t) = \frac{1}{T} e^{-t/T} \quad (19)$$

where the steady-state response is

$$x_{ss} = 0 \quad (20)$$

The error is

$$e(t) = \delta(t) - \frac{1}{T} e^{-t/T} \quad (21)$$

Lastly, the steady-state error is

$$\begin{aligned} e_{ss} &= \lim_{t \rightarrow \infty} e(t) \\ &= 0 \end{aligned} \quad (22)$$

These same solutions can be found in the s-domain where the impulse function is $F(s) = 1$. Consider the s-domain expression solved for $X(s)$,

$$X(s) = \frac{1}{Ts + 1} \quad (23)$$

the steady-state error is shown to be

$$\begin{aligned} x_{ss} &= \lim_{s \rightarrow 0} s \frac{1}{Ts + 1} \\ &= 0 \end{aligned} \quad (24)$$

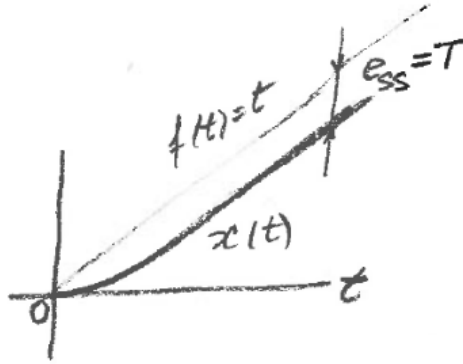
The s-domain error is expressed as

$$E(s) = \frac{T}{Ts + 1} \quad (25)$$

which leads to the steady-state error value

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s \frac{T}{Ts + 1} \\ &= 0 \end{aligned} \quad (26)$$

5.1.3 Ramp response 1st-order system performance indicators



For a 1st-order system subjected to a ramp response, we want to find x_{ss} and e_{ss} . The impulse function is defined as

$$f(t) = t \quad (27)$$

therefore, the response is

$$x(t) = t - T(1 - e^{-t/T}) \quad (28)$$

which leads to the steady-state response

$$x_{ss} = \infty \quad (29)$$

which means that there is no steady-state value. The error is

$$e(t) = T(1 - e^{-t/T}) \quad (30)$$

Lastly, the steady-state error is

$$\begin{aligned} e_{ss} &= \lim_{t \rightarrow \infty} e(t) \\ &= T - \lim_{t \rightarrow \infty} e^{-t/T} \\ &= T \end{aligned} \quad (31)$$

as $\lim_{t \rightarrow \infty} e^{-t/T} = 0$.

These same solutions can be found in the s-domain where the ramp function is $F(s) = \frac{1}{s^2}$. Consider the s-domain expression solved for $X(s)$,

$$X(s) = \frac{1}{Ts + 1} \cdot \frac{1}{s^2} \quad (32)$$

the steady-state error is shown to be

$$\begin{aligned} x_{ss} &= \lim_{s \rightarrow 0} s \frac{1}{Ts + 1} \cdot \frac{1}{s^2} \\ &= \lim_{s \rightarrow 0} \frac{1}{Ts + 1} \cdot \frac{1}{s} \\ &= \infty \end{aligned} \quad (33)$$

therefore, there is no steady-state value. The s-domain error is expressed as

$$\begin{aligned} E(s) &= \frac{Ts}{Ts + 1} \cdot \frac{1}{s^2} \\ &= \frac{T}{Ts + 1} \cdot \frac{1}{s} \end{aligned} \quad (34)$$

which leads to the steady-state error value

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s \frac{T}{Ts + 1} \cdot \frac{1}{s} \\ &= \lim_{s \rightarrow 0} \frac{T}{Ts + 1} \\ &= T \end{aligned} \quad (35)$$

5.2 1st-order System Specific Performance Indicators

Specific performance indicators exist. They depend on system order and excitation type. Examples are:

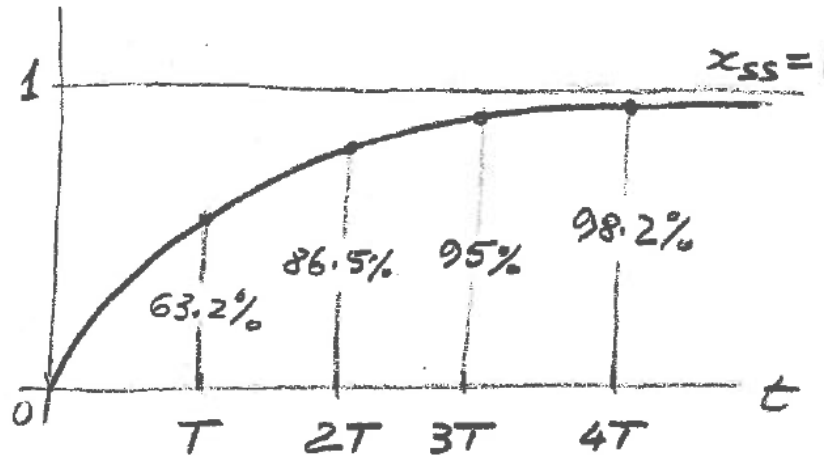
- rise-time $\rightarrow t_r$
- delay time $\rightarrow t_d$
- settling time $\rightarrow t_s$

- decay time / half-time $\rightarrow t_{1/2}$

For a step response, consider the system displacement

$$x(t) = 1 - e^{-t/T} \quad (36)$$

that is plotted as



The rise-time (t_r) of a system

- to rise 63.2% of x_{ss} is $t_{r,63.2\%} = T$
- to rise 86.5% of x_{ss} is $t_{r,86.5\%} = 2T$
- to rise 95% of x_{ss} is $t_{r,95\%} = 3T$
- to rise 98.2% of x_{ss} is $t_{r,98.2\%} = 4T$

in general, $1 - e^{-t/T} = x \rightarrow t = -T \ln(1 - x)$

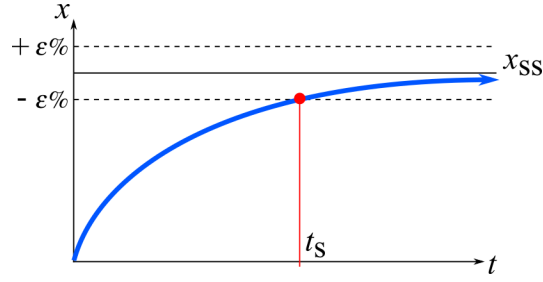
The delay time (t_d) is the time it takes to rise to 50% of x_{ss} ,

$$\begin{aligned} x(t) &= 1 - e^{-t/T} \\ &= 0.5 \end{aligned} \quad (37)$$

therefore, solving for $t = t_d$,

$$\begin{aligned} e^{-t_d/T} &= 0.5 \\ \frac{-t_d}{T} &= \ln(0.5) \\ t_d &= -T \ln(0.5) \\ t_d &= 0.693T \\ t_d &\approx 0.7T \end{aligned} \quad (38)$$

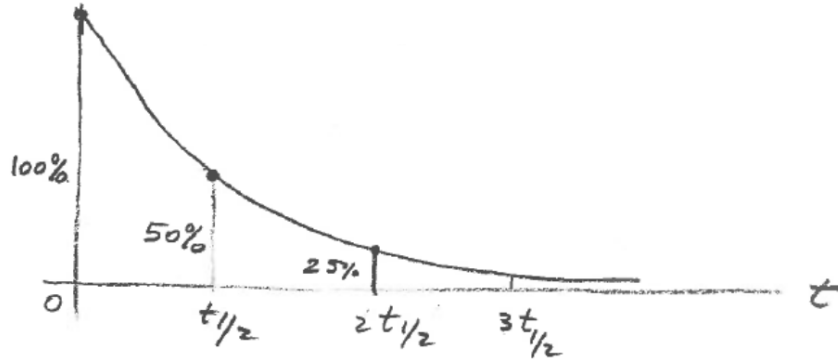
The settling time (t_s) is the time it takes x to get within $\varepsilon\%$ of x_{ss} .



A more precise estimator is the rise time to any selected x which can be defined as

$$\begin{aligned} x_{ss}(1 - e^{-t/T}) &= x & (39) \\ e^{(-t/T)} &= 1 - \frac{x}{x_{ss}} \\ -t/T &= \log \left[1 - \frac{x}{x_{ss}} \right] \\ t &= -T \log \left[1 - \frac{x}{x_{ss}} \right] \end{aligned}$$

5.2.1 Impulse Response Specific Performance Indicators



The half-cycle decay time ($t_{1/2}$) is an impulse response specific performance indicator. Given that the system response to an impulse is

$$x(t) = \frac{1}{T} e^{-t/T} \quad (40)$$

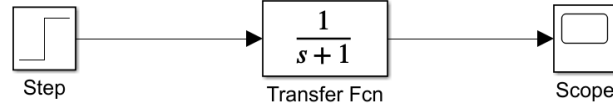
the half-life decay time is where $e^{-t/T} = \frac{1}{2}$, therefore

$$\begin{aligned} t_{1/2} &= -\ln \left(\frac{1}{2} \right) T \\ &\approx 0.693T \end{aligned} \quad (41)$$

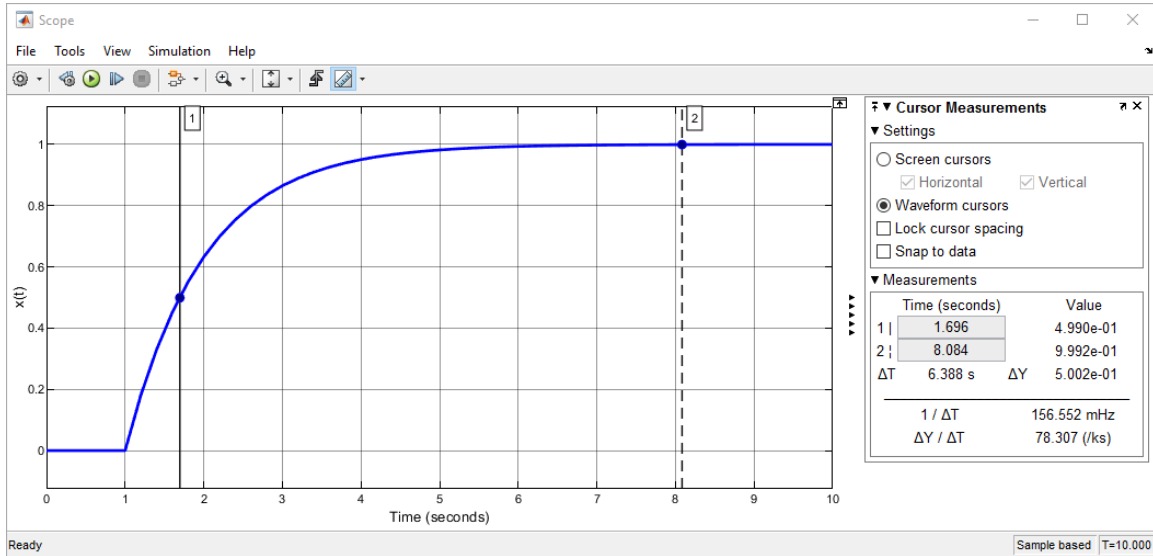
Note that the signal continues to decay by half of its value after each additional $t_{1/2}$.

Example 5.1 SIMULINK Tutorial on Measuring Performance Indicators

Build the simple 1st-order system as shown below



Open the scope and press the ‘Cursor Measurements’ button to activate the cursors.



To measure the delay time t_d , place the first cursor around the point where ‘Value’ measurement is closest to 0.5. $x(t) = 0.5$. Read the ‘Time’ value. This is estimate for t_d . It gives the value $t_d = 0.70$ sec, as the step function happens at 1 sec.

The second cursor can be used to get the settling time t_s . We are going to use the 1% definition of t_d . This means that the response should be around 0.99, or 9.9e-1. Reading the corresponding time value, we get $t_s \approx 7.0$ sec.

5.3 2nd-order System Generic Performance Indicators

Again, starting at the equation of motion

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \omega_n^2f(t) \quad (42)$$

and transfer function of a 2nd-order system

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2} \quad (43)$$

we can expand on this to show

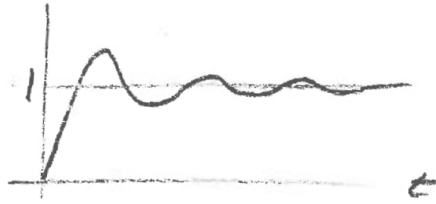
$$\begin{aligned} X(s) &= G(s)F(s) \\ &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} F(s) \end{aligned} \quad (44)$$

next, the error in the s-domain is shown to be

$$\begin{aligned} E(s) &= F(s) - X(s) \\ &= F(s) - G(s)F(s) \\ &= (1 - G(s))F(s) \\ &= \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} F(s) \end{aligned} \quad (45)$$

Again, these are general terms for a 2nd-order system. Note that the definitions for the steady-state value x_{ss} and steady-state error e_{ss} remain largely unchanged from the 1st-order system.

5.3.1 Step response 2nd-order system performance indicators



For a 2nd-order system subjected to a step response, we want to find x_{ss} and e_{ss} . Using the transfer function method to solve for the response, we know the system response is

$$x(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (46)$$

where

$$\omega_d = \omega_n \sqrt{1-\zeta^2} \quad (47)$$

and

$$\phi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} = \sin^{-1} \sqrt{1-\zeta^2} \quad (48)$$

next, the steady-state displacement can be solved for as

$$\begin{aligned} x_{ss} &= \lim_{t \rightarrow \infty} x(t) \\ &= 1 - \frac{\sqrt{1-\zeta^2}}{\zeta} \lim_{t \rightarrow \infty} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \\ &= 1 \end{aligned} \quad (49)$$

as $e^{-\zeta\omega_n t}$ goes to 0. Next, the error as a function of time is

$$\begin{aligned} e(t) &= f(t) - x(t) \\ &= \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \end{aligned} \quad (50)$$

while the steady-state error is

$$\begin{aligned} e_{ss} &= \lim_{t \rightarrow \infty} e(t) \\ &= \frac{\sqrt{1-\zeta^2}}{\zeta} \lim_{t \rightarrow \infty} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \\ &= 0 \end{aligned} \quad (51)$$

as $e^{-\zeta\omega_n t}$ goes to 0.

These same solutions can be found in the s-domain where the step function is $F(s) = \frac{1}{s}$. Consider the s-domain expression solved for $X(s)$,

$$X(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s} \quad (52)$$

the steady-state displacement is shown to be

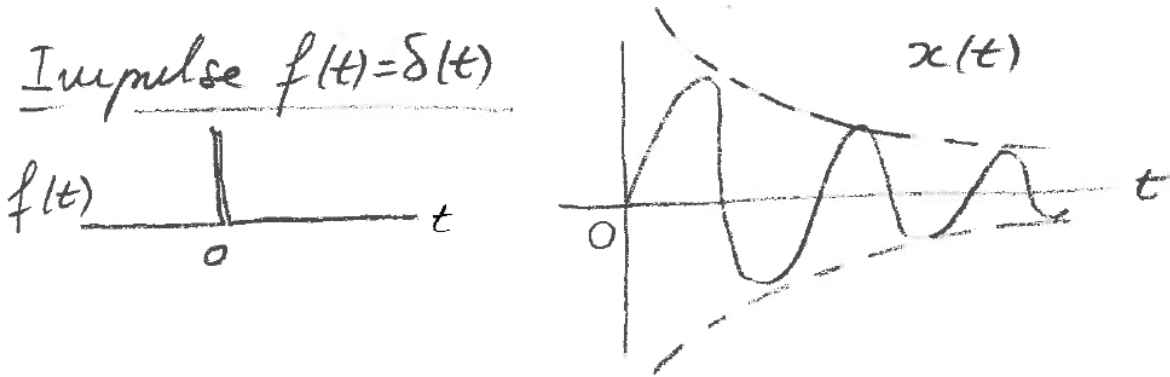
$$\begin{aligned} x_{ss} &= \lim_{s \rightarrow 0} s \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s} \\ &= \frac{\omega_n^2}{\omega_n^2} \\ &= 1 \end{aligned} \quad (53)$$

Next, we can build the s-domain representation of the error as

$$E(s) = \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s} \quad (54)$$

solving for the steady-state error results in

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s} \\ &= \frac{0}{\omega_n} \\ &= 0 \end{aligned} \quad (55)$$

5.3.2 Impulse response 2nd-order system performance indicators

For a 2nd-order system subjected to a impulse response, we want to find x_{ss} and e_{ss} . The impulse function is defined as

$$f(t) = \delta(t) \quad (56)$$

therefore, the response is

$$x(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t) \quad (57)$$

where the steady-state response is

$$\begin{aligned} x_{ss} &= \lim_{t \rightarrow \infty} x(t) \\ &= \frac{\omega_n}{\sqrt{1-\zeta^2}} \lim_{t \rightarrow \infty} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \\ &= 0 \end{aligned} \quad (58)$$

as $e^{-\zeta\omega_n t}$ goes to 0. Next, the error as a function of time is

$$\begin{aligned} e(t) &= f(t) - x(t) \\ &= \delta(t) - \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \end{aligned} \quad (59)$$

Lastly, the steady-state error is

$$\begin{aligned} e_{ss} &= \lim_{t \rightarrow \infty} e(t) \\ &= \lim_{t \rightarrow \infty} \delta(t) - \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \\ &= 0 \end{aligned} \quad (60)$$

as $\delta(t)$ and $e^{-\zeta\omega_n t}$ both go to 0.

These same solutions can be found in the s-domain where the impulse function is $F(s) = 1$. Consider the s-domain expression solved for $X(s)$,

$$X(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (61)$$

the steady-state error is shown to be

$$\begin{aligned} x_{ss} &= \lim_{s \rightarrow 0} s \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ &= 0 \end{aligned} \quad (62)$$

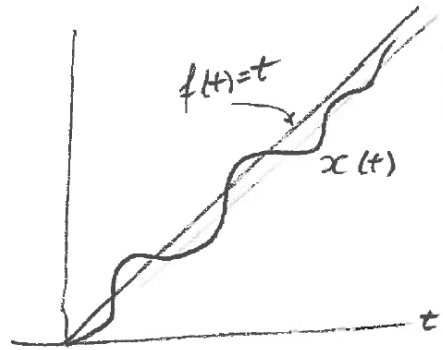
The s-domain error is expressed as

$$E(s) = \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (63)$$

which leads to the steady-state error value

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} s \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ &= 0 \end{aligned} \quad (64)$$

5.3.3 Ramp response 2nd-order system performance indicators



For a 2nd-order system subjected to a ramp response, we want to find x_{ss} and e_{ss} . The impulse function is defined as

$$f(t) = t \quad (65)$$

therefore, the response can be found through the transfer function approach as

$$x(t) = t - \frac{2\zeta}{\omega_n} \left[1 + \frac{1}{\sin \phi_1} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi_1) \right] \quad (66)$$

where

$$\phi_1 = \tan^{-1} \frac{2\zeta\sqrt{1-\zeta^2}}{1-2\zeta^2} \quad (67)$$

which leads to the response

$$\begin{aligned} x_{ss} &= \lim_{t \rightarrow \infty} x(t) \\ &= t - \frac{2\zeta}{\omega_n} \\ &= \infty \end{aligned} \quad (68)$$

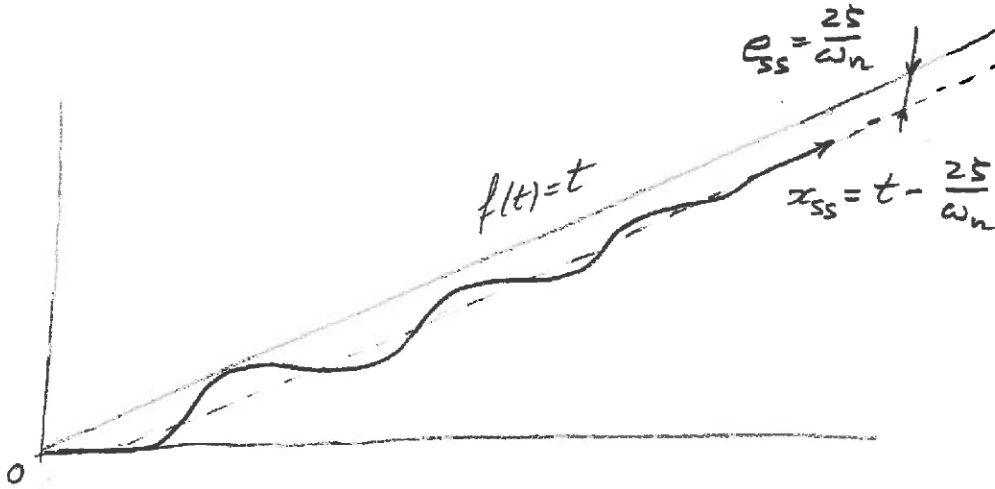
which means that there is no steady-state value. Next, the error as a function of time is

$$\begin{aligned}
 e(t) &= f(t) - x(t) \\
 &= t - \left[t - \frac{2\zeta}{\omega_n} \left[1 + \frac{1}{\sin \phi_1} e^{-\zeta \omega_n t} \sin(\omega_d t + \phi_1) \right] \right] \\
 &= \frac{2\zeta}{\omega_n} \left[1 + \frac{1}{\sin \phi_1} e^{-\zeta \omega_n t} \sin(\omega_d t + \phi_1) \right]
 \end{aligned} \tag{69}$$

Lastly, the steady-state error is

$$\begin{aligned}
 e_{ss} &= \lim_{t \rightarrow \infty} e(t) \\
 &= \frac{2\zeta}{\omega_n} \lim_{t \rightarrow \infty} \left[1 + \frac{1}{\sin \phi_1} e^{-\zeta \omega_n t} \sin(\omega_d t + \phi_1) \right] \\
 &= \frac{2\zeta}{\omega_n}
 \end{aligned} \tag{70}$$

as the term in the brackets goes to 1.



These same solutions can be found in the s-domain where the ramp function is $F(s) = \frac{1}{s^2}$. Consider the s-domain expression solved for $X(s)$,

$$X(s) = \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2} \cdot \frac{1}{s^2} \tag{71}$$

the steady-state error is shown to be

$$\begin{aligned}
 x_{ss} &= \lim_{s \rightarrow 0} s \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s^2} \\
 &= \lim_{s \rightarrow 0} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s} \\
 &= \lim_{s \rightarrow 0} \frac{1}{s} \\
 &= \infty
 \end{aligned} \tag{72}$$

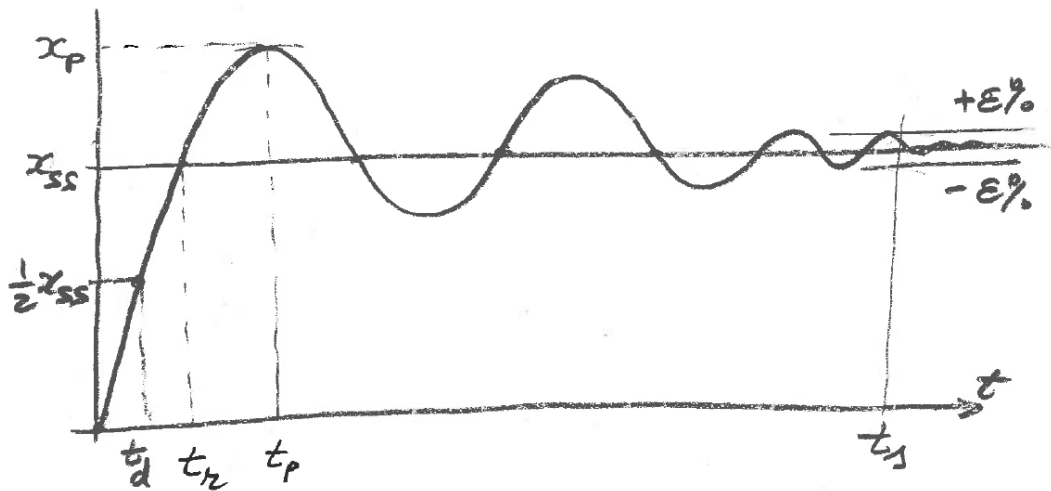
therefore, there is no steady-state value. The s-domain error is expressed as

$$E(s) = \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s^2} \tag{73}$$

which leads to the steady-state error value

$$\begin{aligned}
 e_{ss} &= \lim_{s \rightarrow 0} s \frac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s^2} \\
 &= \lim_{s \rightarrow 0} \frac{s^2(s + 2\zeta\omega_n)}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s^2} \\
 &= \frac{2\zeta\omega_n}{\omega_n^2} \\
 &= \frac{2\zeta}{\omega_n}
 \end{aligned} \tag{74}$$

5.4 2nd-order System Specific Performance Indicators



Specific performance indicators exist. They depend on system order and excitation type. Examples are:

- rise time $\rightarrow t_r$
- peak time $\rightarrow t_p$
- peak value $\rightarrow x_p$
- settling time $\rightarrow t_s$
- delay time $\rightarrow t_d$
- max percentage overshoot $\rightarrow M_p$

Many of these are the same as a 1st-order system or taken directly from the system response. The max percentage overshoot (M_p) is defined as

$$M_p = \left(\frac{x_p}{x_{ss}} - 1 \right) \cdot 100 \quad (75)$$

5.4.1 Procedure for a Step Response

Given the 2nd-order system, with the transfer function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (76)$$

where ω_n is the natural frequency and ζ is the critical damping ratio. Considering that the system is subjected to a step response, we know the system response is

$$x(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (77)$$

where we can calculate:

$$\omega_d = \omega_n \sqrt{1-\zeta^2} \quad (78)$$

and

$$\phi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} = \sin^{-1} \sqrt{1-\zeta^2} \quad (79)$$

Next, compute the values for the performance indicators.

First the rise time is calculated by the fact that $x(0) = 0$ and $x(t_r) = 1$. From equation 77, the mean height of the system is shown to be when $\sin(\omega_d t + \phi) = 0$, therefore, as $\sin(\pi) = 0$, we can show that $\omega_d t + \phi = \pi$, rearranging this yields

$$t_r = \frac{\pi - \phi}{\omega_d} \quad (80)$$

To find the peak time (t_p), we need to find the peak, which is defined as $\frac{dx}{dt} \Big|_{t=t_p} = 0$, or drawn as



where we consider only the decaying part of the signal caused by the step function, or $x = e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$. Therefore

$$\begin{aligned}
 0 &= \frac{dx}{dt} \\
 &= \frac{d}{dt} \left[e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \right] \\
 &= -\zeta\omega_n e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) + \omega_d e^{-\zeta\omega_n t} \cos(\omega_d t + \phi) \\
 \zeta\omega_n \sin(\omega_d t + \phi) &= \omega_d \cos(\omega_d t + \phi) \\
 \frac{\sin(\omega_d t + \phi)}{\cos(\omega_d t + \phi)} &= \frac{\omega_d}{\zeta\omega_n}
 \end{aligned} \tag{81}$$

By converting the left hand side of this equation to $\tan(\omega_d t_p + \phi)$ yields

$$\begin{aligned}
 \tan(\omega_d t_p + \phi) &= \frac{\omega_d}{\zeta\omega_n} \\
 &= \frac{\sqrt{1-\zeta^2}}{\zeta} \\
 &= \tan(\phi)
 \end{aligned} \tag{82}$$

when considering the definition of $\tan$ provided by equation 79. We must find t_p such that $\tan(\omega_d t_p + \phi) = \tan(\phi)$. Given that the function $\tan(\alpha)$ repeats itself after π , 2π , 3π , $\dots$ as shown in figure 5.1, we know $\tan(\alpha) = \tan(\alpha + \pi) = \tan(\alpha + 2\pi), \dots$.

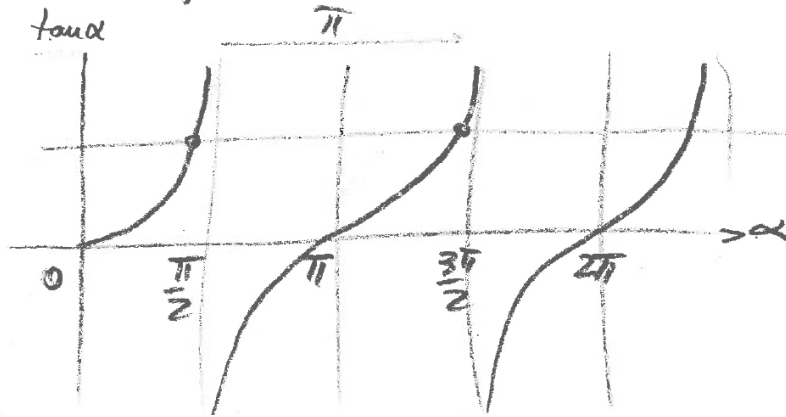


Figure 5.1: Plotting the $\tan(\alpha)$

Hence, picking back up from equation 82 lead to

$$\begin{aligned}
 \omega_d t_p + \phi &= \phi + \pi \\
 \omega_d t_p &= \pi
 \end{aligned} \tag{83}$$

which results in

$$t_p = \frac{\pi}{\omega_d} \quad (84)$$

The peak value (x_p) can be found as

$$\begin{aligned} x_p &= x(t_p) \\ &= 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n \frac{\pi}{\omega_d}} \sin(\omega_d(\pi/\omega_d) + \phi) \\ &= 1 + \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n \frac{\pi}{\omega_d}} \sin(\phi) \\ &= 1 + \frac{1}{\sqrt{1-\zeta^2}} \left(e^{-\frac{\zeta}{\sqrt{1-\zeta^2}} \pi} \right) \sqrt{1-\zeta^2} \\ &= 1 + e^{-\frac{\zeta}{\sqrt{1-\zeta^2}} \pi} \end{aligned} \quad (85)$$

when considering that $\sin(\phi + \pi) = -\sin(\phi)$ and $\sin(\phi) = \sqrt{1-\zeta^2}$.

The max overshoot (M_p) can be found as

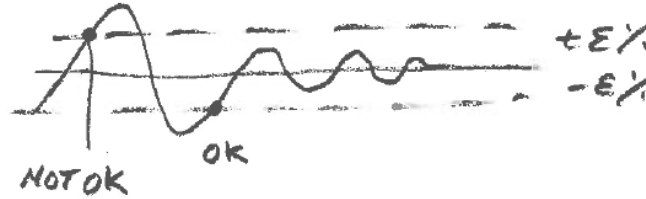
$$\begin{aligned} M_p &= \frac{x_p - x_{ss}}{x_{ss}} \\ &= \frac{1 + e^{-\frac{\zeta}{\sqrt{1-\zeta^2}} \pi} - 1}{1} \\ &= e^{-\frac{\zeta}{\sqrt{1-\zeta^2}} \pi} \end{aligned} \quad (86)$$

for a few select damping ratios, the overshoot percentages are shown in Table 1. However, the typical range of damping is $0.4 < \zeta < 0.8$ and therefore the typical max overshoot is $0.25\% < M_p < 1.5\%$.

Table 1: Overshoot percentages for select damping ratios.

ζ	0	0.2	0.4	0.6	0.8	1
M_p	100.00%	52.68%	25.40%	9.49%	1.52%	0.00%

The definition of settling time (t_s) is to “get withing $\pm\epsilon\%$ of x_{ss} and stay so”.



The settling time is defined as the time when both

$$|x_{ss} - x(t_s)| < \Delta \quad (87)$$

and

$$|x_{ss} - x(t > t_s)| < \Delta \quad (88)$$

are true where $\Delta = \varepsilon \cdot x_{ss}$. This is shown in figure 5.2.

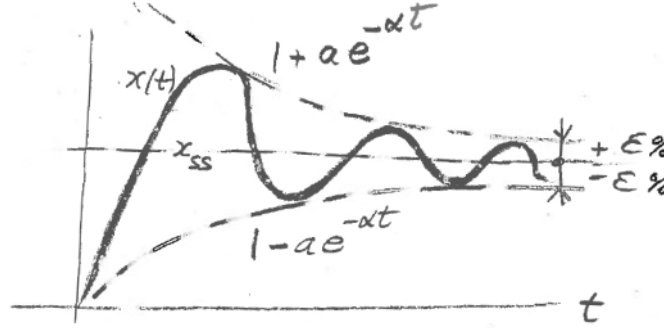


Figure 5.2: Settling time for a 2nd-order system.

We can find a generalized expression for t_s if we consider the system response for $x_{ss} = 1$ as

$$x(t) = 1 - \left[\frac{1}{\sqrt{1 - \zeta^2}} \right] e^{-[\zeta \omega_n]t} \sin(\omega_d t + \phi) \quad (89)$$

redefining the items in the brackets as a and α , respectively, the system response can be simplified to

$$x(t) = 1 - ae^{-\alpha t} \sin(\omega_d t + \phi) \quad (90)$$

Therefore, the envelop of the system response is $1 \pm ae^{-\alpha t}$ while the settling condition is $ae^{-\alpha t} = \Delta$. Considering that the final peak above the error range will happen when $a \cong 1$, we make the approximate calculation

$$a = \frac{1}{\sqrt{1 - \zeta^2}} \Big|_{\zeta < 1} \cong 1 \quad (91)$$

and considering that t_s is when the system decays under the value ε , we need to find the t value when

$$e^{-\alpha t} = \varepsilon \quad (92)$$

therefore, setting $\varepsilon = 2\%$, we can find

$$e^{-\alpha t} = 0.02 \quad (93)$$

$$-\alpha t = \log(0.02) \quad (94)$$

$$= -3.9$$

$$\cong -4$$

Therefore, knowing that $\alpha = \zeta \omega_n$, we can deduce

$$-\alpha t_s \cong -4 \quad (95)$$

$$-\zeta \omega_n t_s \cong -4$$

$$t_s \cong \frac{4}{\zeta \omega_n}$$

when $\zeta \ll 1$ this simplified expression is within $\pm 2\%$.

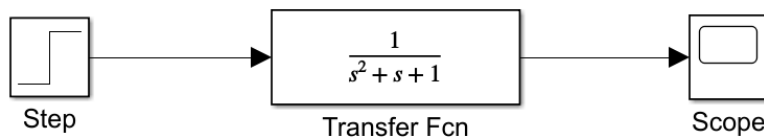
Importantly, one should always consider the effects of ζ and ω_n on performance.

- An increase in ω_n shortens rise time (t_r), peak time (t_p), and settling time (t_s).
- An increase in ζ reduces max overshoot (M_p) and shortens settling time (t_s).

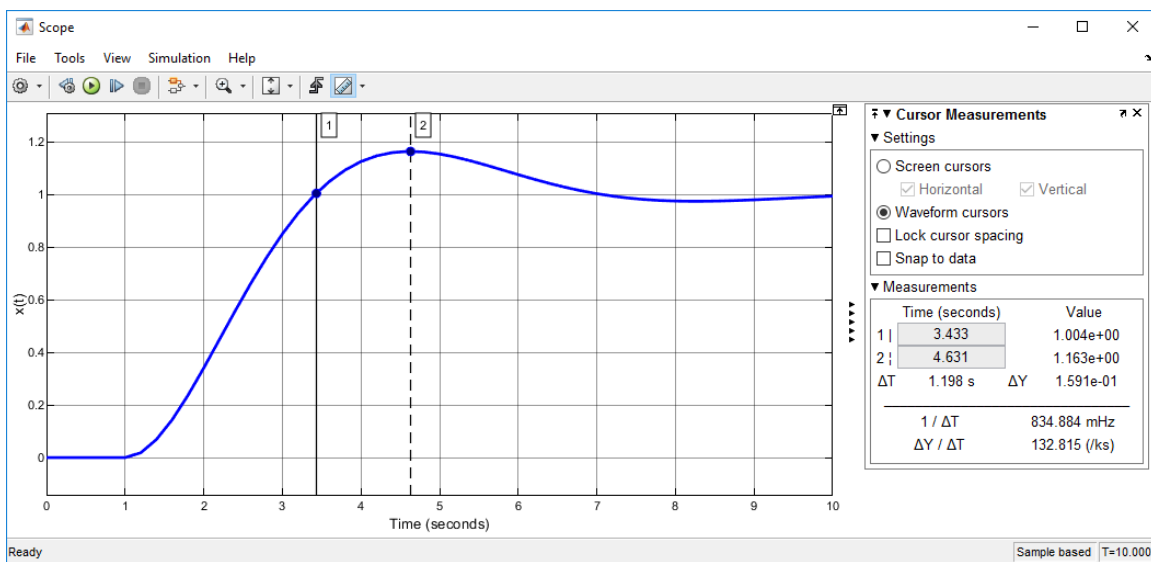
As before, the definition of time delay (t_d) for a 2nd-order system is the time it takes to rise to 50% of x_{ss} the first time. Similarly, the max percentage overshoot (M_p) for a 2nd-order system is calculated in the same way as a 1st-order system.

Example 5.2 SIMULINK Tutorial on Measuring Performance Indicators

Build the simple 2nd-order system as shown below



Open the scope and press the 'Cursor Measurements' button to activate the cursors.



Use the first cursor to find the first crossing of $x_{ss} = 1$. Read the time as $t_r = 2.433$ sec as the step function starts at 1 sec. Place the second cursor at peak value. Read the peak time $t_p = 3.631$ sec and peak amplitude $x_p = 1.163$. Calculate $M_p = 16.3\%$.